

Automated Medical Insurance Claim Denial & Appeals Agent

MediAppeal AI — IBM watsonx Multi-Agent System

Programme	Edunet Foundation IBM SkillsBuild Training
Domain	Healthcare FinTech / AI-Powered Enterprise Automation
Platform	IBM watsonx Orchestrate + IBM Granite-3.2-8b
Submitted by	PS-38 Final Project Submission

■ THE CHALLENGE

Healthcare administrative staff and patients waste millions of hours fighting complex health insurance claim denials. Reviewing dense insurance policy rules, matching them to clinical records, and drafting valid medical appeal letters manually is a massive bureaucratic roadblock. Because the process is slow and requires deep legal and medical knowledge, many valid claims are abandoned, leading to severe financial losses for hospitals and unexpected out-of-pocket medical debt for patients.

Scale of the problem: U.S. providers lose an estimated **\$8.6 billion annually** in uncontested claim denials. Approximately **56%** of denied claims are never resubmitted, and a skilled medical billing specialist spends an average of **73 minutes** manually reviewing a single denial before drafting an appeal.

■ THE OBJECTIVE

Build an **AI-powered Multi-Agent Medical Insurance Claim Denial & Appeals System** that automates the end-to-end audit, evidence gathering, and appeal generation pipeline — reducing a 73-minute manual workflow to milliseconds with zero human intervention on qualifying cases.

■ CORE PILLARS

1. Insurance Policy Retrieval (RAG)

Ingest policy frameworks (ACC-901, ORTHO-404, ONCO-202) to flag exact clause violations and map each denial code to its governing coverage criteria.

2. Clinical Evidence Extraction

Securely scan unstructured medical charts using rule-based NLP to pull proving biometrics — EKG findings, PT duration logs, companion diagnostic results — and produce an evidence verdict with confidence scoring.

3. Automated Appeal Generation

Draft legally structured appeal letters with clause citations, clinical justification blocks, and forecasted approval probability ratings. Route unwinnable cases to manual review with a detailed deficiency report.

■ SPECIALIZED AGENTS

Agent	Role	Output
Policy Audit Agent	Maps denial code → governing policy rule	Rule ID, criteria, excerpt
Clinical Data Agent	NLP scan of clinical notes for evidence	criteria_met, confidence, rationale
Appeal Strategy Agent	Drafts appeal letter or review notice	Disposition + full letter text
Success Predictor Agent	Forecasts approval probability	Probability % (e.g. 94%)

■ MANDATORY TECH STACK

IBM Langflow / IBM watsonx Orchestrate + IBM Models (Granite-3.2-8b)

Layer	Technology	Purpose
AI Engine	IBM Granite-3.2-8b-instruct	Language understanding & generation
Orchestration	IBM watsonx Orchestrate / Langflow	Multi-agent pipeline coordination
Backend	Python 3.13 · Flask 3.1 · Pandas 3.0	REST API server & data processing
Frontend	Bootstrap 5.3 · Chart.js 4.4	Responsive web dashboard
Schema	OpenAPI 3.0.3	watsonx Orchestrate skill registration
Data	CSV denial records · Policy TXT	Patient records & rule document store

■ PROJECT DELIVERABLES

- **app.py** — Flask server with 4 routes: /, /process, /add-case, /chat, /skills
- **data/insurance_policies.txt** — NHCAB policy rules document (ACC-901, ORTHO-404, ONCO-202)
- **data/denial_records.csv** — 10 realistic patient denial case records
- **config/orchestrate_skill_schema.json** — OpenAPI 3.0.3 watsonx Orchestrate skill schema
- **templates/index.html** — Premium IBM-palette web dashboard with live console, analytics, and chat
- **templates/skills.html** — Skill catalog viewer with full JSON schema display
- **README.md** — Full project report with architecture diagrams and execution metrics